Information For the Use
Piperacillin (4000mg) + Tazobactum (500mg)

Piprazon 4.5 Injection

Piperacillin (4000mg) + Tazobactam (500mg) Injection: Comprehensive Information

Generic Name: Piperacillin + Tazobactam
Strength: 4000mg of Piperacillin and 500mg of Tazobactam
Formulation: Injectable Solution
Drug Class: Antibiotic Combination

Description

Piperacillin + Tazobactam Injection is a combination of two antibiotics: **Piperacillin**, a broad-spectrum penicillin antibiotic, and **Tazobactam**, a beta-lactamase inhibitor. This combination works synergistically to treat a wide range of bacterial infections. Piperacillin works by inhibiting bacterial cell wall synthesis, leading to bacterial death, while Tazobactam protects Piperacillin from degradation by bacterial beta-lactamases, which would otherwise render Piperacillin ineffective. This enhanced activity makes the combination particularly effective against resistant bacterial strains.

Indications and Uses

Piperacillin + Tazobactam Injection is used for the treatment of infections caused by susceptible bacteria, including:

1. **Respiratory Tract Infections:**
   - **Community-acquired pneumonia (CAP)** and **hospital-acquired pneumonia (HAP)**.
   - **Acute exacerbations of chronic obstructive pulmonary disease (COPD)**.
   - **Sinusitis** and **bronchitis**.
2. **Urinary Tract Infections (UTIs):**
   - **Complicated UTIs**, including **pyelonephritis** and **cystitis**.
3. **Intra-abdominal Infections:**
   - **Peritonitis**, **appendicitis**, and other intra-abdominal infections, including those with abscess formation.
4. **Skin and Soft Tissue Infections:**
   - **Cellulitis**, **abscesses**, and **wound infections**.
5. **Bone and Joint Infections:**
   - **Osteomyelitis** and **septic arthritis**.
6. **Gynecological Infections:**
   - **Pelvic inflammatory disease (PID)** and **endometritis**.
7. **Bloodstream Infections (Sepsis):**
   - **Bacteremia** and **septicemia** caused by susceptible organisms.
8. **Other Serious Infections:**
   - **Endocarditis** and **meningitis** in certain clinical situations.

Mechanism of Action

- **Piperacillin:** A **penicillin-class antibiotic** that inhibits bacterial cell wall synthesis by binding to **penicillin-binding proteins (PBPs)**. This weakens the bacterial cell wall and leads to cell lysis and death. It is effective against a wide range of Gram-positive and Gram-negative bacteria, including some **anaerobes**.
- **Tazobactam:** A **beta-lactamase inhibitor** that prevents the breakdown of Piperacillin by bacterial enzymes called **beta-lactamases**. By inhibiting these enzymes, Tazobactam allows Piperacillin to remain effective against resistant bacterial strains.

Dosage and Administration

- **Route of Administration:**
   - Piperacillin + Tazobactam is administered **intravenously (IV)**. It can be given either through **intermittent infusion** or **continuous infusion**,

depending on the clinical situation and severity of the infection.

- **Recommended Dosage:**
  - The typical adult dose is **4g of Piperacillin** and **500mg of Tazobactam**, administered every **6–8 hours** based on the severity and type of infection.
  - **For severe infections (e.g., sepsis, pneumonia):** Doses may be higher or given more frequently, as determined by a healthcare provider.
  - **Renal impairment:** Dose adjustments are required in patients with impaired kidney function. A dose reduction or extended dosing interval may be necessary.
- **Administration Instructions:**
  - The injection should be given over a **30-minute to 1-hour period** depending on the specific administration protocol.
  - **Reconstitution and dilution**: The solution should be prepared according to the manufacturer's guidelines, and only sterile water or saline should be used for reconstitution.
  - **Monitoring**: Vital signs and lab parameters (e.g., kidney and liver function) should be monitored during treatment.

Precautions Before Use

Do not use Piperacillin + Tazobactam Injection if you:

- Are allergic to Piperacillin, Tazobactam, or any other **penicillin**-type antibiotics.
- Have a **history of severe allergic reactions** (anaphylaxis) to beta-lactam antibiotics.
- Are pregnant or breastfeeding, unless specifically prescribed by a healthcare professional, as it can pass into breast milk.
- Have a **history of liver disease** or **jaundice**, as this combination may exacerbate these conditions.

Use with caution in patients with:

- **Renal impairment** (dose adjustments are required in patients with kidney problems).
- **Gastrointestinal conditions**, particularly **colitis**, as antibiotics can increase the risk of **Clostridium difficile-associated diarrhea**.
- **Electrolyte imbalances** or **hypertension**, especially if administered with other nephrotoxic or antihypertensive agents.
- **Blood disorders**, such as **anemia** or **thrombocytopenia**.

Potential Side Effects

1. **Common Side Effects:**
   - **Injection site reactions** (pain, swelling, redness).
   - **Gastrointestinal symptoms**, such as **nausea**, **vomiting**, **diarrhea**, and **abdominal discomfort**.
   - **Rash** or **pruritus** (itching).
   - **Headache** and **fever**.
2. **Serious Side Effects:**
   - **Severe allergic reactions** (e.g., anaphylaxis, difficulty breathing, swelling of the throat or face).
   - **Clostridium difficile-associated diarrhea (CDAD)**, a potentially life-threatening condition.
   - **Renal toxicity**, especially in patients with preexisting kidney conditions.
   - **Hepatic dysfunction**: Elevated liver enzymes, jaundice, or liver failure in rare cases.
   - **Blood dyscrasias**: Low blood cell counts, such as **leukopenia** (low white blood cells) or **thrombocytopenia** (low platelets).
3. **Long-Term Use Side Effects:**
   - **Superinfections** due to the destruction of normal bacterial flora, leading to resistant bacterial growth.
   - **Electrolyte imbalances** and **hypokalemia** (low potassium), particularly when used in combination with other drugs that affect kidney function.

Drug Interactions

1. **Medications to Avoid or Monitor:**
    - **Probenecid**: Can increase the blood concentration of Piperacillin, potentially leading to toxicity.
    - **Anticoagulants** (e.g., warfarin): Piperacillin + Tazobactam may increase the effects of anticoagulants, increasing bleeding risk.
    - **Aminoglycosides**: When used concurrently, there is an increased risk of kidney toxicity.
    - **Diuretics**: Certain diuretics may exacerbate kidney damage when combined with Piperacillin + Tazobactam.
2. **Laboratory Test Interactions:**
    - May cause **false-positive** results on **urine glucose tests**.
    - Monitoring of **renal function** and **liver enzymes** is recommended during therapy.

Overdose and Emergency Management

Symptoms of Overdose:

- **Severe gastrointestinal upset**, such as nausea, vomiting, and diarrhea.
- **Electrolyte imbalances**, especially related to **renal failure**.
- **Signs of allergic reactions** such as rash, swelling, or difficulty breathing.

Management:

- **Supportive care** is the mainstay of treatment for overdose.
- **Hydration** and **electrolyte correction** may be needed.
- In cases of severe allergic reactions, administer **epinephrine** or **antihistamines** immediately.

Storage Instructions

- Store at **room temperature** (20–25°C / 68–77°F), away from direct light and moisture.
- **Do not freeze** the injectable solution.
- Keep the medication out of reach of children.

Warnings and Legal Considerations

- **Pregnancy Category B**: Piperacillin + Tazobactam has been shown to be safe during pregnancy when prescribed by a healthcare provider, although it should be used only if necessary.
- **Controlled Substance**: This medication is not a controlled substance but should be used only as prescribed to avoid complications.

Manufactured in India for:

Cafoli™ LIFECARE

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